

Static Rigid Bodies

Concurrent Forces

Concurrent forces refer to forces that pass through a common point. If the body is in static equilibrium, the sum of forces is 0, hence the sum of their components is 0.

Thus,

$$F_x = F_y = F_z = 0$$

Principle of Transmissibility

Even if the forces do not originate from the same point, principle of transmissibility states that the forces can be slid along their line of action for easier calculations.

This also means that the converse is true, if a body is in static equilibrium and has arbitrary forces applied to it, the sum of the forces is 0.

Non-Concurrent Forces

If the forces are non-concurrent, the forces give rise to moments in the rigid body.

Thus, for static equilibrium, the sum of moments about any point is 0.

Hence we can use the equation to determine unknown forces in a non-concurrent force system:

$$R_x = R_y = M_o^R = 0$$

or as one force and two moment equations, where the force equation is perpendicular to the line formed by AB:

$$R_x = 0 \text{ or } R_y = 0, M_A^R = M_B^R = 0$$

or 3 moment equations, where the points selected are not collinear:

$$M_A^R = M_B^R = M_C^R = 0$$

Free Body Diagrams

im gonna kill myself!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

Reactions and Supports

Recall the different types of reactions from MECH1:

Rollers and Frictionless Surface Contacts: single force normal to surface (known direction)

Cords/Strings and short linkages: single force in line with the string/linkage (known direction)

collar on frictionless rod/sliding pin in slot: single force perpendicular to travel direction (known direction)

Pin joints/Friction surface contact: single force with unknown direction (2 components)

Fixed Supports: single force with unknown direction and couple